

36104 · ASSESSMENT 1

Critique and Repair

See what a chart claims · repair it with the Visual Vocabulary · verify a machine's reading of its data

Weight	30% of the subject · individual
Covers	Sessions 1–2: <i>Seeing Data</i> and <i>Choosing Visual Forms</i>
Released	Seeing Data session — Thursday 30 July 2026
Supervised component	25-minute AI-restricted window at the start of the Dashboards-session studio — Thursday 13 August 2026
Due	Monday 24 August 2026, 11:59 pm (Sydney time)
Submit	One completed <code>A1_template.ipynb</code> · your data file(s) in <code>data/</code> · a PNG/screenshot of the original chart

The task

You will take one published chart from the wild and work it through the course pipeline: see what it claims, *repair* it with the Visual Vocabulary, and *verify* a machine's interpretation of its data. The first two sessions rehearse every part of this: the Seeing Data studio (Critique and Repair sheet, Critique 1) rehearses Parts A and C; the Choosing Visual Forms studio (Redesign ×3, Chart Choice Decision Record) rehearses Part B.

Part A — Seeing (the critique)

What does the chart show, and what does it claim? Selection, encoding, framing, omission — keeping the discipline from Seeing Data: *directly visible vs interpretation rather than observation*. Ends with a repaired caption. **Drafted by hand in the supervised, AI-restricted window on 13 August**, then transcribed into the notebook's `critique` cell.

Part B — Choosing (the repair)

Obtain or faithfully reconstruct the chart's underlying data. Produce **three redesigns**: your *best form* — the chart the original should have been — plus **two alternatives serving different Visual Vocabulary tasks**. Defend the best form with a completed **Chart Choice Decision Record** (the same form as the Choosing Visual Forms studio, including the rejected alternative and the specific reason). The Decision Record is what is marked; the alternatives prove you considered the space.

Part C — Verifying (the claim audit)

Generate an interpretation of your dataset using the **supplied prompt** (verbatim, in the template). Classify **every claim** with the four-way taxonomy from the Seeing Data lab:

Label	Meaning
supported	the data directly backs it — attach an evidence probe
plausible but unverified	sounds right; this dataset cannot settle it
unsupported	asserted with no evidence in this data either way
contradicted	the data shows otherwise — attach an evidence probe

Probes are zero-argument functions returning the evidence (a filtered or aggregated DataFrame/Series) — required for every `supported` and `contradicted` claim. Finish with the five-question AI disclosure.

Your artefact

Choose one artefact from the **released, verified pool** or propose your own found-in-the-wild chart by the end of the Seeing Data session: a real published chart, a named publisher and date, and an obtainable or reasonably reconstructable data source. Approval is required. Two students may study the same source chart, but every submission must be individual and independently produced.

Data rule. If the exact data is not published, reconstruct it faithfully (digitise, or synthesise to match the chart's visible quantities) and say so: set `data_status = "reconstructed"` and write the reconstruction note. This is the atlas book's own method — representative data, never passed off as the original measurement. Undisclosed reconstruction is an academic-integrity issue. Disclosed reconstruction remains eligible for full marks; its fidelity and limitations are judged under H2.

Copyright note. Include only the chart image needed for criticism and review, cite its publisher, creator (where known), date and source URL, and do not republish the assessment beyond the subject site. Follow any additional UTS library or copyright guidance supplied with the task.

AI policy

Part A is drafted AI-free in the supervised window. An assistant may be used in Part B under the notebook contract: docstrings first, accept or reject completions, and nothing counts until the checks pass. Part C requires an assistant-generated interpretation; students who cannot or do not wish to use an assistant can request a course-supplied generated interpretation. The disclosure block is assessed: report use accurately, including legitimate non-use in Part B. False or misleading disclosure is an academic-integrity issue.

Submission contract (read this twice)

The template notebook defines named variables and functions (`META`, `critique`, `original_data`, `redesign_best` ... `disclosure`). The autograder executes your notebook top-to-bottom on a clean machine and inspects those names. **Run the self-check cell before submitting** — it is the same code the marker runs. A notebook that does not execute end-to-end scores 0 on the automated items until resubmitted (one resubmission, capped at 80%).

Rubric — 100 marks (scaled to 30% of the subject)

Automated (60 marks, scored by `a1_autograder.py`)

ID	Item	Marks	Check
G0	Notebook executes end-to-end	6	all cells run clean, no errors
G1	Metadata complete and valid	6	<code>META</code> fields present; URL well-formed; <code>data_status</code> $\in$ {obtained, reconstructed}
G2	Data provenance	10	<code>original_data</code> loads; ≥ 6 rows $\times$ ≥ 2 columns; no empty columns; reconstruction note ≥ 30 words when reconstructed
G3	Redesign mechanics	16	three Axes returned; three distinct Vocabulary categories, all valid; bar-family charts include a zero baseline; every chart titled with ≥ 15 characters
G4	Claim-audit structure	14	interpretation ≥ 400 chars from the supplied prompt; ≥ 6 claims; every label from the four-way taxonomy; ≥ 1 <code>supported</code> , ≥ 1 <code>plausible but unverified</code> , ≥ 1 <code>unsupported</code> or <code>contradicted</code> ; an executable, non-empty probe for every <code>supported</code> / <code>contradicted</code> claim
G5	Critique + Decision Record structure	4	all 9 critique fields (incl. repaired caption) 5–150 words; all 9 Decision Record fields present, defence fields ≥ 10 words
G6	Five-question AI disclosure	4	all 5 answers completed, none placeholder
Total		60	

Human-marked (40 marks, anchored 0/1/2 per item)

Markers read the autograder report first, then judge only these five items.

Item (weight)	0	1	2
H1 · Critique insight (×5)	Restates the chart	Correctly identifies the claim and one real omission	Also separates <i>directly visible</i> from <i>interpretation</i> , names the framing device, and the repaired caption actually repairs
H2 · Reconstruction fidelity (×2.5)	Data unrelated to the original's quantities	Plausible magnitudes and units	Matches the original's visible values, or deviations explicitly disclosed
H3 · Decision Record defence (×5)	No argument, or category label only	Correct task category with a generic justification	Ties audience → task → channel precision; the rejected alternative is real and the reason specific
H4 · Redesign craft (×5)	Unreadable or misleading	Honest defaults, readable	Titles state findings; scales honest; labels carry the argument
H5 · Audit judgement (×2.5)	Classifications mostly wrong	Right labels, thin probes	Probes genuinely discriminate — a contradicted probe could have exonerated the claim; plausible but unverified used precisely, not as a dodge
Total: 40 marks (weights × anchor score; maximum $2 \times (5 + 2.5 + 5 + 5 + 2.5) = 40$)			

Process requirements (not marked, but gates)

- Part A supervised draft handed in during the 13 August window (paper or photo). Missing draft → H1 capped at 1.
- Artefact selected from the released pool, or approval recorded for bring-your-own.
- One resubmission permitted for G0 failures, capped at 80% overall.

What good looks like

A strong A1 reads like the class notebooks: the critique names the claim in one sentence and the repaired caption says what the original's should have said; the Decision Record argues from the reader's task rather than from taste, and its rejected alternative is one you genuinely considered; the reconstruction note is boringly specific; at least one probe shows a fluent-sounding claim to be **contradicted**; and the disclosure says exactly what the assistant drafted, what you rejected, and how you caught its mistakes.

Questions: raise them in the studio or contact the subject coordinator. AI use in this assessment is governed by the AI policy above; unauthorised use is academic misconduct under UTS rules.